Name the force and its source

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify forces on a chosen object and name each interaction partner.

SUCCESS CRITERIA - I can...

1. Name the chosen object and each force source.
2. Distinguish mass from weight with units.
3. Include only forces acting on the chosen object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “interaction” and “weight” to explain your answer. Sentence starters are optional.

interaction: two objects affecting one another

weight: the gravitational force on an object

normal force: a support force perpendicular to a contact surface

01 Warm-up: bring back an earlier idea

Velocity changes from **-4** to **-24 m/s** in **4 s**, then stays at **-24 m/s** for **8 s**. **Find acceleration** and **describe the speed trend** in each interval.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_g = mg$.
Use g = **9.8 N/kg** unless stated. Normal force is perpendicular to contact; tension follows the string. **Name the source**.

Worked reasoning: A book rests on a table. Earth pulls the book down; the table pushes the book up. Each force label names both the force and the object producing it.

Your turn: 1. **Draw the book's force diagram**.
2. **Name the force** on Earth paired with the book's weight.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Add and label force arrows.

Book resting on a level table; draw forces on the book only.

Marzano check - circle: ① ② ③

03 Inventory the interactions

A rope pulls a **5 kg** crate to the right across a rough level floor. The crate slides right; the rope is horizontal.
1. **Draw every external force** on the crate and **name its source**.
2. Which magnitudes can you find using g = **9.8 N/kg**?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Add and label force arrows.

The dot represents the object. Chosen object: crate

Marzano check - circle: ① ② ③

04 Mass and weight in two places

A **3.0 kg** instrument is moved from Earth to the Moon. Use g_Earth = **9.8 N/kg** and g_Moon = **1.6 N/kg**. **Complete the comparison table** and **explain why mass** and weight behave differently.

Sentence starter: Both ... . They differ in ... because ... .

Location	Mass (kg)	Weight (N)
Earth	3	____
Moon	3	____

Marzano check - circle: ① ② ③

Name the force and its source

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Normal is not always weight

A rope helps support a **5 kg** box on a level floor. Its upward force component is **12 N**, and the box has no vertical acceleration. Use g = **9.8 N/kg**. **Complete the vertical force diagram** and **find the normal force**. **Explain why it** is less than mg.

Sentence starter: Both … . They differ in … because … .



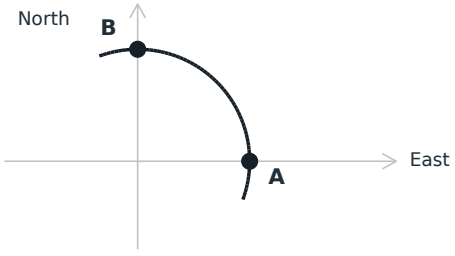
Add the floor's normal force separately.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A puck follows a circle at a steady **5 m/s**. A student calls it equilibrium. **Draw velocity** at two points and correct the statement.
Use the curve for the turning case; **show how velocity changes**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Path sketch: runner moves from A toward B.

Marzano check - circle: ① ② ③

07 Exit check

A **2.0 kg** object hangs at rest from a vertical string.
1. **Name the two forces**, their sources, magnitudes and directions.
2. Is object-on-string a force on the object?

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I notice … . This supports … because … .

Add and label force arrows.

Hanging object at rest: label sources and magnitudes.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **2 kg** book rests on a table while a hand pushes down with **15 N**.
1. Use g=**10 N/kg**.
2. **Find the normal force** and **draw the third-law partner of** each force on its separate object.

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Book pressed down by a hand: draw forces on the book.

Marzano check - circle: ① ② ③